Do Now · retrieval

Quicksort: pivot and partition

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

In bubble sort, which value reaches its final position at the end of pass one?

Q2

What is a base case?

Q3

Which element is a[3] in [6,2,9,4,5]?

Q4

What is the result of 4 <= 6 and why?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Pick a pivot — Use the rightmost element, 6. A left boundary i tracks where the next small value belongs.

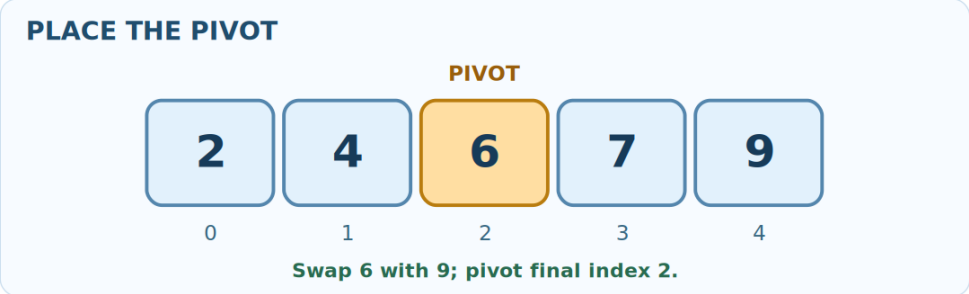
Core idea 2

Scan and move — Only values less than or equal to 6 move left. We move 2, then 4, without sorting those two values.

Core idea 3

Place the pivot — Swap 6 into index 2; elements to the left are <=6, and those to the right are >6. Each side still needs sorting.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

pivot — Element selected to split a quicksort partition.

partition — Rearrange values so smaller/equal values are on one side of the pivot.

Lomuto — A partition approach scanning left to right and placing the pivot last.

index — Zero-based position within the array.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Partition [6,3,8,2,5] around final-element pivot 5.

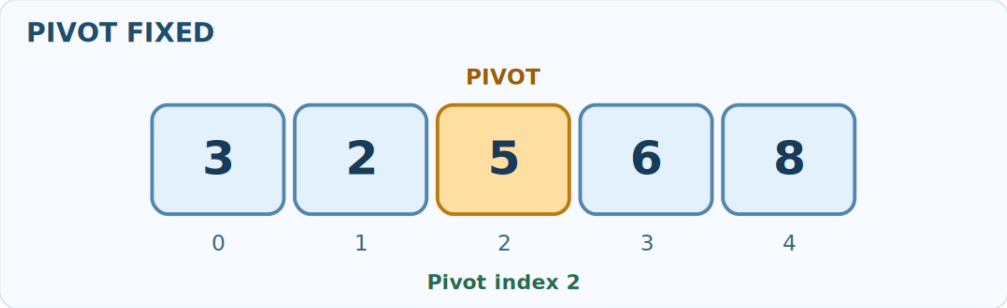
Follow the worked method

1. Predict the next state — 6 is larger than 5; 3 is smaller and swaps into the left boundary.

2. Reveal and explain — 8 is larger; 2 swaps to index 1, giving [3,2,8,6,5].

3. Reveal and explain — Swap pivot 5 with position 2: [3,2,5,6,8]. The two sides are not yet necessarily sorted.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict the next state
Compare 8 and 1 with pivot 6. Move 1 into index 0.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Partition [4,1,3] using the final value as pivot, with Lomuto partition. Give the result and pivot index.

Question 2

Partition [9,2,5,1,6] using the final pivot. Show the array after moving each qualifying value and after placing the pivot.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain why choosing the last element as pivot can lead to $O(n^2)$ work on already sorted data, and suggest an improvement.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A quicksort partition automatically fully sorts the two sides of the pivot.

Use key vocabulary from memory

1. Explain pivot in your own words.

2. Explain partition in your own words.

3. Explain Lomuto in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.